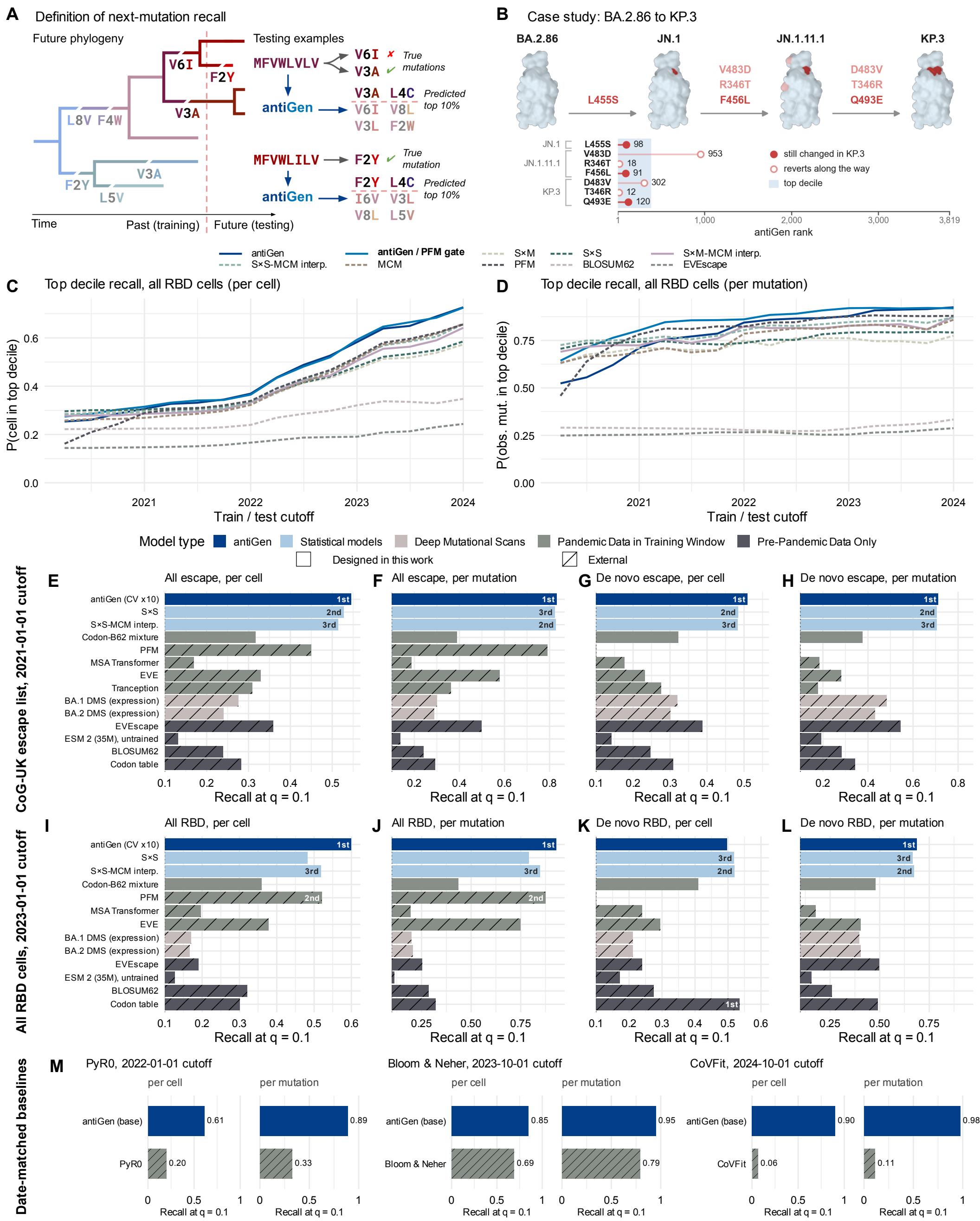

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
Time
Past (training)
Future (testing)
B
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
R346T
F456L
D483V
T346R
Q493E
JN.1
JN.1.11.1
KP.3
L455S 98
V483D 953
R346T 18
F456L 91
D483V 302
T346R 12
Q493E 120
still changed in KP.3
reverts along the way
top decile
1
1,000
2,000
3,000
3,819
antiGen rank
antiGen
S×S-MCM interp.
antiGen / PFM gate
MCM
S×M
PFM
S×S
BLOSUM62
S×M-MCM interp.
EVEscape
C
Top decile recall, all RBD cells (per cell)
P(cell in top decile)
0.0
0.2
0.4
0.6
2021
2022
2023
2024
Train / test cutoff
D
Top decile recall, all RBD cells (per mutation)
P(obs. mut. in top decile)
0.00
0.25
0.50
0.75
2021
2022
2023
2024
Train / test cutoff
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Designed in this work
External
CoG-UK escape list, 2021-01-01 cutoff
All RBD cells, 2023-01-01 cutoff
Date-matched baselines
E
All escape, per cell
F
All escape, per mutation
G
De novo escape, per cell
H
De novo escape, per mutation
antiGen (CV x10)
S×S
S×S-MCM interp.
Codon-B62 mixture
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.6
0.8
Recall at q = 0.1
I
All RBD, per cell
J
All RBD, per mutation
K
De novo RBD, per cell
L
De novo RBD, per mutation
M
PyR0, 2022-01-01 cutoff
Bloom & Neher, 2023-10-01 cutoff
CoVFit, 2024-10-01 cutoff
per cell
per mutation
antiGen (base)
PyR0
0.61
0.89
0.20
0.33
Bloom & Neher
0.85
0.95
0.69
0.79
CoVFit
0.90
0.98
0.06
0.11
0
0.5
1
Recall at q = 0.1